

Methods of Reducing Friction.

- 1 Polishing: polishing the surface, reduces the irregularities and hence, reduces friction.
- 2 Lubrication: it provides a layer of the lubricant on which the bodies can easily slide.
- 3 use of ball bearing or roller bearing: Roller bearing helps to convert sliding friction into a milder form - rolling friction. Rolling friction is lesser than sliding friction.
- 4 Streamlining: Fast cars, boats, planes, etc., have a streamlined body. This is to allow air (or water in the case of boats) to easily flow by, without offering much resistance.
- 5 use of correct combination of surfaces in contact: use of alloys on moving and sliding parts ~~reducing~~ reduces friction because alloys have a low coefficient of friction.

What is Friction?

Friction is the force acting b/w two surfaces. It is a resistance that acts b/w two moving objects resisting motion.

Friction is essential for offering traction which facilitates motion. All vehicles depend upon friction for speeding up, slowing down and even for changing

directions.

Our day to day activities, like walking to your favourite book store to piling dishes on top of each other, friction is what makes all that possible. Burning meteor before they hit the earth's surface and causing forest fires, friction can have many forms.

comparison between advantages and disadvantages of friction.

Advantages

- Ability to walk
- Piling of objects
- stopping of moving vehicles
- Lights up a fire
- Transfer of energy
- writing
- wear accessories
- gripping on to things
- Generating heat
- Burning of meteor

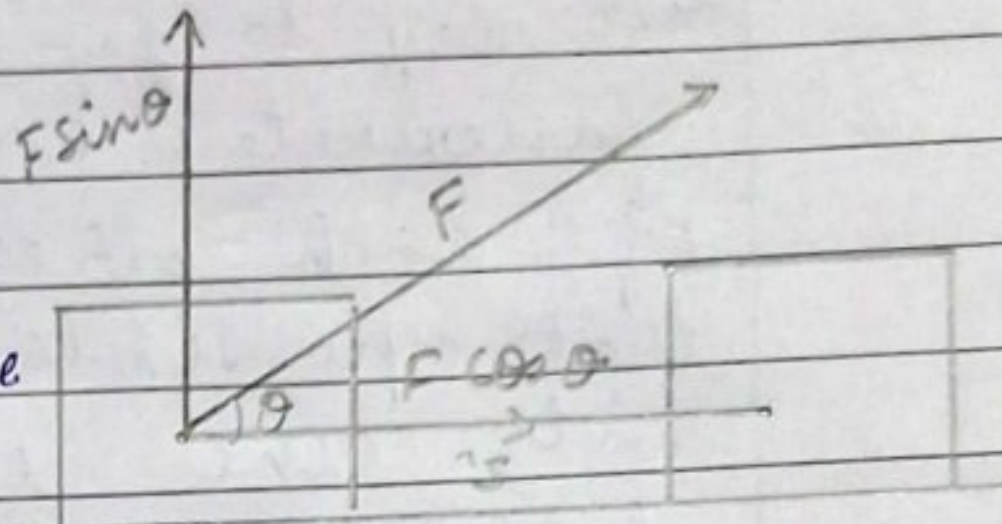
Disadvantages.

- loss of energy.
- Resists motion.
- Decreases life of moving machinery parts.
- overheating
- Forest fires.
- wear and Tear.
- Decreased efficiency
- excessive noise production.

$$F = \frac{mv}{\Delta t}$$

Work, Energy and Power.

work: work is said to be done when some force is applied on the body and the body moves through some distance in the direction of applied force.



$$W = F \cos \theta \cdot S$$

$$W = FS \cos \theta$$

* If θ is acute ($< 90^\circ$).
 $\cos \theta$ is positive.
 W is positive.

e.g. If $\theta = 0^\circ$.

$$W = FS \cos 0$$

$$W = FS \text{ Positive work.}$$

* If θ is obtuse ($> 90^\circ$).
 $\cos \theta$ is negative.
 W is negative.

e.g. If $\theta = 180^\circ$.

$$W = FS \cos 180$$

$$W = FS (-1)$$

$$W = -FS \text{ Negative work.}$$

example: (1) work done by friction is negative work.

(2) work done by gravity on a particle or a particle thrown upwards.

$$\begin{array}{cccc} 0 & \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{\sqrt{3}}{2} & 1 \\ 60^\circ & 45^\circ & 30^\circ & 0^\circ & \end{array}$$

*

If $\theta = 90^\circ$.

$$W = FS \cos 90^\circ$$

$$W = FS (0)$$

$$W = 0 \text{ zero work.}$$

examples.

①

work done by a coolie is zero.

②

work done by centripetal force is zero.

$$W = FS \cos \theta$$

$$W = \vec{F} \cdot \vec{S}$$

$$[\vec{A} \cdot \vec{B} = AB \cos \theta]$$

clearly work done is a scalar quantity.
In terms of rectangular components.

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\vec{S} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$W = \vec{F} \cdot \vec{S}$$

$$W = (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) \cdot (x \hat{i} + y \hat{j} + z \hat{k})$$

$$W = xF_x + yF_y + zF_z$$

Q.

A force of $(2\hat{i} - \hat{j} + 3\hat{k})\text{N}$ acts on a particle and the particle moves through a displacement of $(3\hat{i} + 2\hat{j} + \hat{k})\text{m}$. Find the work done.

$$W = FS \cos \theta$$

$$W = \vec{F} \cdot \vec{S}$$

$$W = (2\hat{i} - \hat{j} + 3\hat{k}) \cdot (3\hat{i} + 2\hat{j} + \hat{k})$$

$$W = 6 - 2 + 3 = 6 + 1 = 7\text{J}$$

Q.

A force of 10N moves a body through a distance of 5m. Find the work done if force makes an angle of 60° .

$$W = FS \cos \theta$$

$$W = 10 \times 5 \times \cos 60^\circ = 10 \times 5 \times \frac{1}{2} = 25\text{J}$$



SI unit of work.
= Nm = Joule.

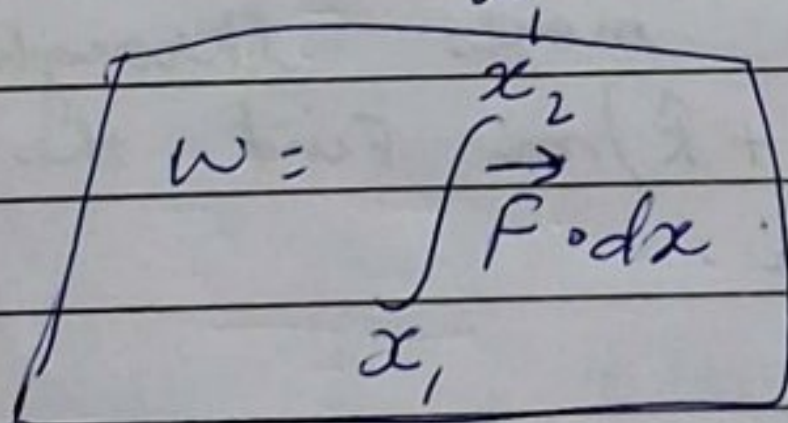
C.G.S unit of work.
= dyne cm = erg.

inf Work done by a variable force.
If at any instant $\vec{F}$ is the force acting on a body and $d\vec{x}$ the body moves through a small displacement $d\vec{x}$ then, the small work done is given by

$$dw = \vec{F} \cdot d\vec{x}.$$

Total work done in moving the body from $x = x_1$ to $x = x_2$.

$$fdw = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x}.$$


$$W = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x}$$

Q. A body moves from $x = 1\text{m}$ to $x = 4\text{m}$ under the application of force given by $F = (x^3 + 2x)\text{ N}$. Find the work done if the force and the displacement are in the same direction.

we know, $W = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x}$

$$\begin{array}{r} 316 \\ 12 \\ \hline 96 \\ 16x \\ \hline 56 \end{array}$$

—/—/—

$$W = \int_1^4 (x^3 + 2x) dx \cos 0^\circ$$

$$W = \int_1^4 x^2 dx \cos 0^\circ$$

$$W = \int_1^4 (x^3 + 2x) dx \cos 0^\circ$$

$$W = \int_1^4 (x^3 + 2x) dx$$

$$W = \int_1^4 x^3 dx + \int_1^4 2x dx$$

$$W = \left(\frac{x^4}{4} \right)_1^4 + 2 \left(\frac{x^2}{2} \right)_1^4$$

$$W = \frac{1}{4} (x^4)_1^4 + \frac{2}{2} (x^2)_1^4$$

$$W = \frac{1}{4} (4^4 - 1^4) + (4^2 - 1^2)$$

$$W = \frac{1}{4} (256 - 1) + (16 - 1)$$

$$W = \frac{255}{4} + 15$$

$$W = 63.75 + 15 = 78.75 \text{ J.}$$

POWER.

It is defined as the rate of doing work. That is work done in one second. If 'W' is the work done in time 't' then power is given by $P = \frac{W}{t}$.

Instantaneous power.

If at any instant 'dw' is the work done in small time dt then instantaneous power is given by

$$P = \frac{dw}{dt}$$

v^{imp.}

Expression for power in terms of velocity

$$P = \frac{dw}{dt}$$

$$P = \frac{W}{t} \quad a =$$

$$P = \frac{F \cdot S}{t}$$

$$P = \frac{\vec{F} \cdot d\vec{x}}{dt}$$

$$P = \frac{m \cdot a \cdot S}{t}$$

$$P = m \cdot$$

$$P = \frac{\vec{F} \cdot d\vec{x}}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

Clearly power is a scalar quantity.

$$P = \frac{W}{t}$$

SI unit of power = $\frac{\text{Joule}}{\text{second}} = \text{watt}$.

$$1 \text{ kW} = 10^3 \text{ watt}$$

$$1 \text{ MW} = 10^6 \text{ watt}$$

commercial unit of power = horse power.

$$1 \text{ hp} = 746 \text{ watt}$$

$$\text{D.F. of power} = \frac{M^1 L^2 T^{-2}}{T^1} = M^1 L^2 T^{-3}$$

Energy.

- 1 Kinetic Energy (E): The energy possessed by a body by virtue of its motion is known as kinetic energy.

If 'm' is the mass of the a body and 'v' is its velocity, then K.E of the body is given by,

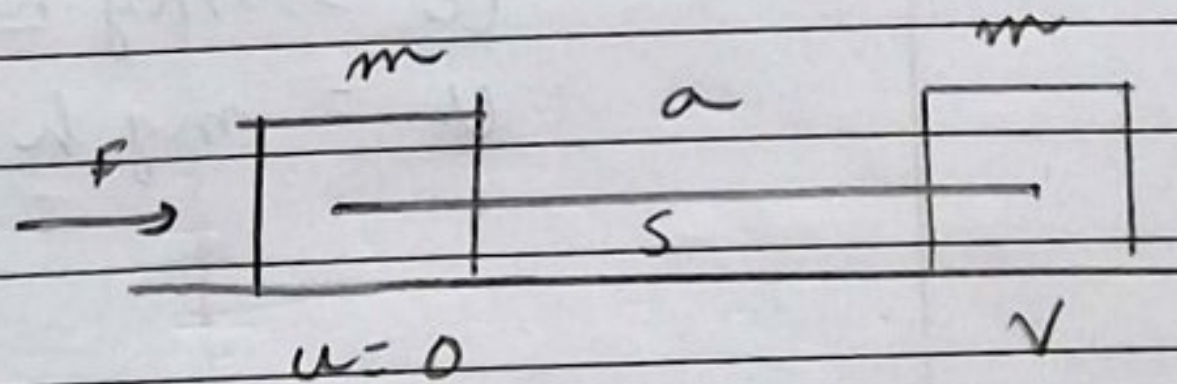
$$E = \frac{1}{2} mv^2$$

Proof:

$$\text{work done} = \text{kinetic E}$$

$$= FS \cos \theta$$

$$W = E = FS \cos 0^\circ = FS$$



$$E = mas \rightarrow (1)$$

using eqn $v^2 - u^2 = 2as$

$$v^2 - 0^2 = 2as$$

$$a = \frac{v^2}{2s} \rightarrow (2)$$

put ② in ①

$$E = m \frac{v^2}{2}$$

$$E = \frac{1}{2} m v^2$$

minimum force to be applied to lift a body in upward direction is equal to its weight (mg).

2 Potential energy (U) : It is the energy possessed by a body by virtue of its position or configuration (i.e. when some conservative force is acting on the body).

Types.

a) Gravitational potential energy.

It is the potential energy possessed by a body in gravitational field. Minimum force to be applied $F = mg$.

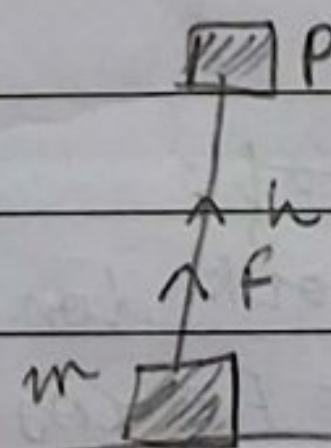
Displacement = h .

Work done = Potential energy = $Fh \cos \theta$.

$$W = U = Fh \cos 0^\circ$$

$$U = mgh \cdot 1$$

$$U = mgh$$



conservation of Mechanical Energy
Mechanical energy of a body is defined as the sum total of its kinetic energy and potential energy when in motion.

conservation of mechanical energy implies that the total mechanical energy of a body remains constant if the forces acting on it are conservative in nature.

conservation of mechanical energy for a freely falling body

At point P

Kinetic energy

$$K_1 = \frac{1}{2} m (0)^2 = 0$$

Potential energy

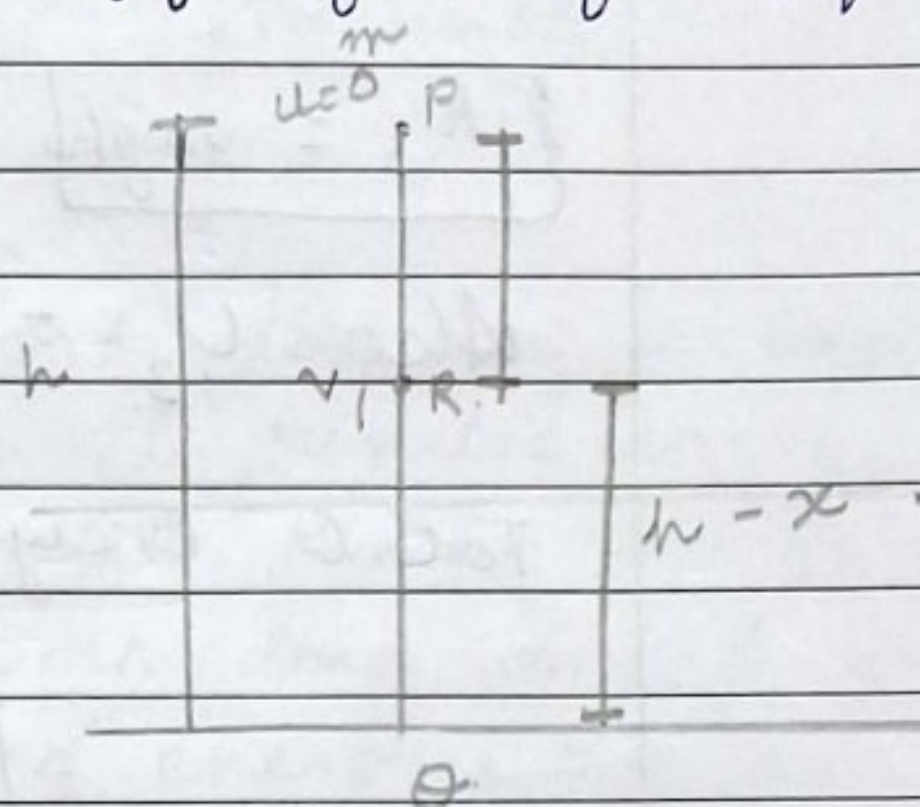
$$U_1 = mgh$$

Total energy at point P

$$E_1 = K_1 + U_1$$

$$E_1 = 0 + mgh$$

$$E_1 = mgh$$



At point R

$$K_2 = \frac{1}{2} m v_1^2$$

we have,

$$v_1^2 - 0^2 = 2gx$$

$$v_1^2 = 2gx$$

$$\therefore K_2 = \frac{1}{2} m \cdot 2gx$$

$$K_2 = mgx$$

$$\begin{aligned} \text{Potential energy} &= U_2 \\ &= mg(h-x) \end{aligned}$$

Total energy

$$E_2 = K_2 + U_2$$

$$E_2 = mgx + mg(h-x)$$

$$E_2 = mgx + mgh - mgx$$

$$E_2 = mgh$$

At point Q.

$$K_3 = \frac{1}{2} m v^2.$$

we have,

$$v^2 - 0^2 = 2gh.$$

$$v^2 = 2gh.$$

$$K_3 = \frac{1}{2} m \cancel{2} gh.$$

$$\boxed{K_3 = mgh}$$

$$\text{Also } U_3 = mg(0) = 0.$$

$$\text{Total energy } E_3 = K_3 + U_3.$$

$$E_3 = mgh + 0.$$

$$\boxed{E_3 = mgh}.$$

clearly, $E_1 = E_2 = E_3$. i.e., mechanical energy of the body remains constant. [conserved].

Q A crane lifts a body of mass 1000 kg to a height of 20 m in 5 seconds.

Find the power of the crane. (take $g = 10 \text{ m/s}^2$)

$$P = \frac{W}{t}.$$

$$\begin{aligned} \text{Now, } W &= mgh = P.E. \\ &= 20 \times 10 \times 1000 = 4 \times 10^4 \text{ watt.} \end{aligned}$$

Q The engine of a train has to apply a force of 4000 N to overcome friction to run at a rate uniform rate of 72 km/hr. what is the power developed by the engine?

$$\begin{aligned}
 1 \text{ km/hr} &= \frac{5}{18} \text{ m/s} \\
 &= \frac{1000 \text{ m}}{60 \times 60 \text{ sec}} = \frac{5}{18} \text{ m/sec}
 \end{aligned}$$

$$F = 4000 \text{ N}$$

$$V = 72 \text{ km/hr}$$

$$V = 72 \times \frac{5}{18} \text{ m/s} = 20 \text{ m/s}$$

$$\text{Now, } P = FV$$

$$P = 4000 \times 20 = 8 \times 10^4 \text{ watt}$$

Work Energy Theorem.

It states that the work done on a body is equal to change in its kinetic energy.

If ' K_i ' is the initial kinetic energy of the body, ' w ' is the work done on it and ' K_f ' is its final kinetic energy, then acc to

$$\text{W.E.T, } w = K_f - K_i$$

$$W = FS \cos 0^\circ$$

$$w = mas$$

$$\text{Also } v^2 - u^2 = 2as$$

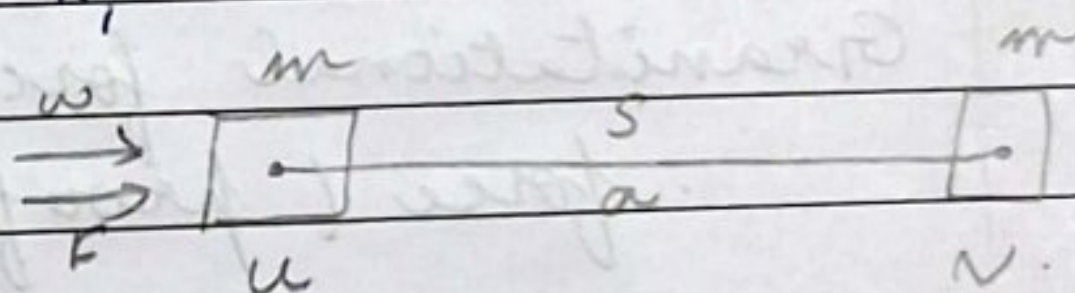
$$\Rightarrow a = \frac{v^2 - u^2}{2s}$$

$$\therefore w = m \left(\frac{v^2 - u^2}{2s} \right) s$$

$$w = \frac{1}{2} m (v^2 - u^2)$$

$$w = \frac{1}{2} mv^2 - \frac{1}{2} mu^2$$

$$\Rightarrow \boxed{w = K_f - K_i}$$



conservative and non-conservative forces.

↓
A force is said to be conservative if no work is done in moving a body around a closed path in the given field.
e.g; gravitational force, electrostatic force.

Another definition:

A force is said to be conservative if work done by it or work done against it is independent of path and dependent only on the initial and final position.

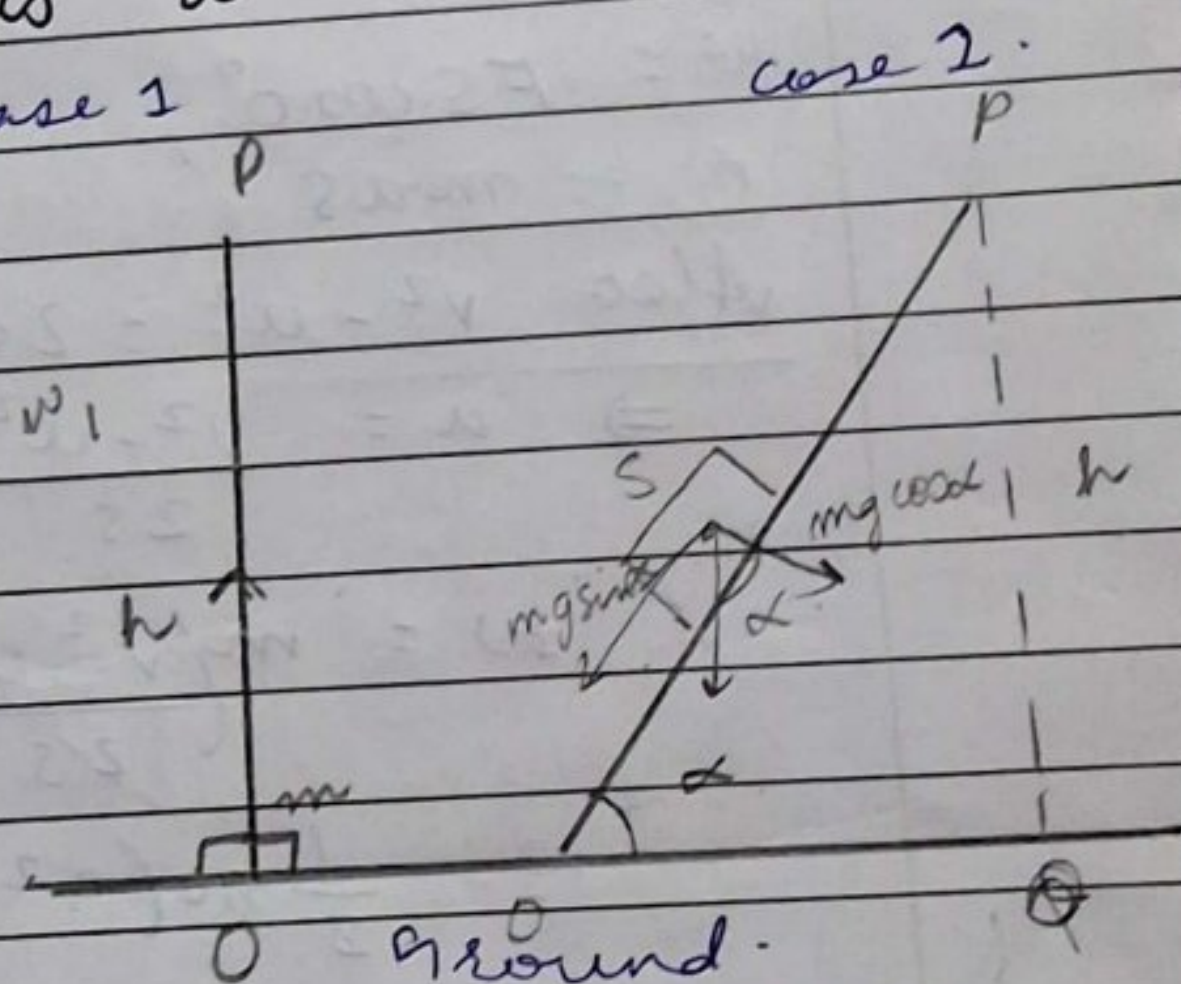
Gravitational force is a conservative force (proof)

Case 1:

$$W_1 = FS \cos \theta$$

$$W_1 = mgh \cos 0^\circ$$

$$W_1 = mgh$$



Case 2:

$$W_2 = FS \cos \theta$$

$$W_2 = mg \sin \alpha \times S \cos 0^\circ$$

$$W_2 = mg \sin \alpha \cdot S$$

$$W_2 = mg \frac{h}{S} \times S$$

$$W_2 = mgh$$

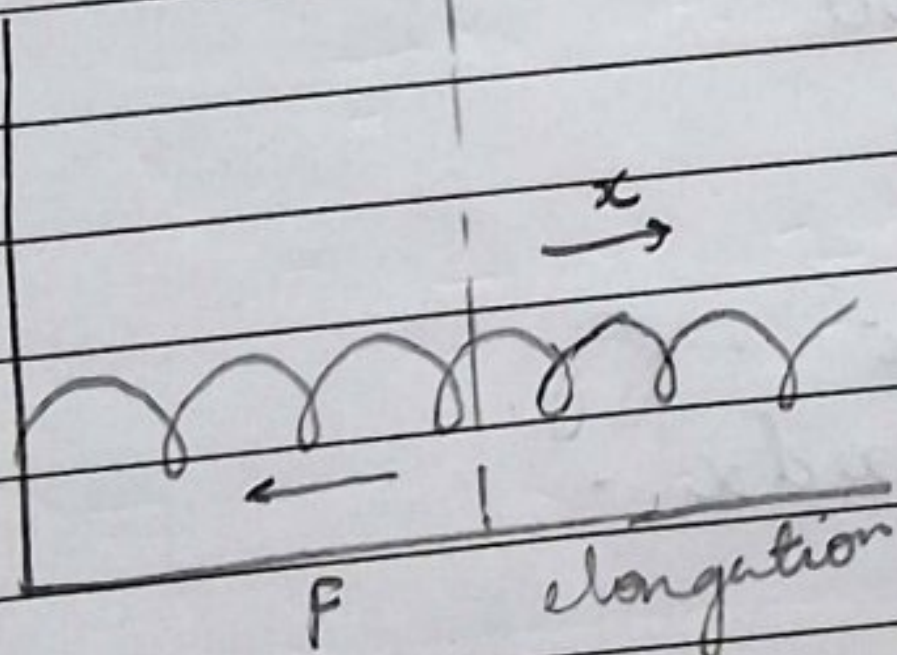
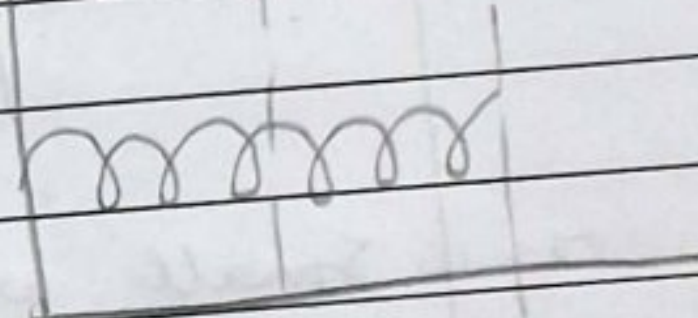
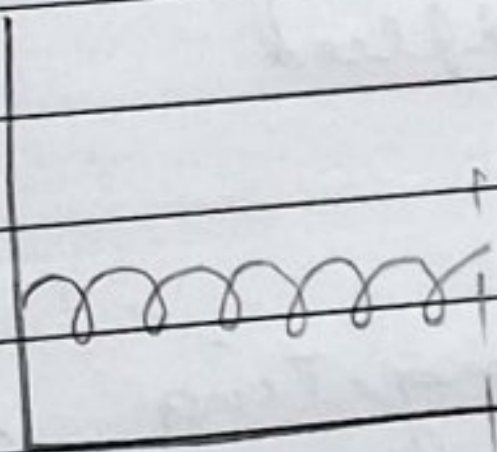
clearly $W_1 = W_2$

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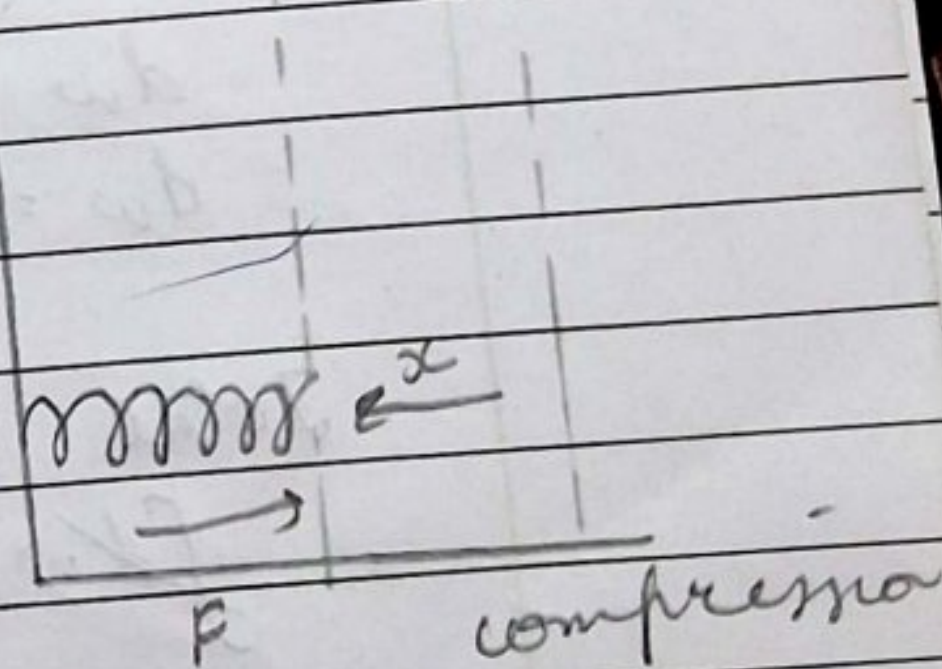
Non conservative force: a force is said to be non conservative when work done by it or ^{work done} against it depends on the path between two points.
e.g; frictional force.

Dissipative force: It is the force due to which energy is lost by a system.
e.g; air resistance and frictional force.

b) Elastic Potential energy of a spring
Restoring force: It is the force developed in ~~the~~ a spring when its elongated or compressed. It is due to this force, due to which a spring restores its original configuration.



$$F \propto x$$



$$F \propto -x$$

$$F \propto -x$$

$$\boxed{F = -kx} \text{ Hook's law.}$$

k is a constant and is known as spring constant.

The restoring force developed in a spring is directly proportional to the elongation or compression produced in it.

The work done in elongating or compressing a spring is stored in it in the form of elastic potential energy.

Minimum force to be applied

$$F = kx.$$

→ Small work done in elongating or compressing a spring by a small amount dx .

$$dw = F dx \cos 0^\circ$$

$$dw = kx dx.$$

Total work done

$$\int dw = \int_0^x kx dx.$$

$$w = k \int_0^x x' dx$$

//_

$$W = k \left(\frac{x^2}{2} \right)^x$$

$$W = \frac{1}{2} k (x^2)^0^x$$

$$W = \frac{1}{2} k (x^2 - 0^2)$$

$$\boxed{W = \frac{1}{2} k x^2}$$

This work done is stored in the spring in the form of elastic potential energy.

$$\therefore U = \frac{1}{2} k x^2.$$